Sion An

Postdoctoral Researcher,
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Research Statement

My research lies at the intersection of Brain-Computer Interfaces (BCI) and Artificial Intelligence. I specialize in developing meta-learning frameworks to address the inherent inter-subject variability in biosignals. By bridging the gap between human neural signals and intelligent systems, I aim to build adaptive BCI systems that facilitate both machine control and personalized clinical interventions.

Keywords: Meta-Learning, Transfer Learning, Medical Image and Signal Processing, Brain-Computer Interfaces.

Research Experience

- 2026-03 – Present · **Postdoctoral Researcher**, Graduate School of Artificial Intelligence, Pohang University of Science and Technology (POSTECH), affiliated with the Medical Artificial Intelligence Lab, Department of Computer Science and Engineering, under the supervision of Prof. Sang Hyun Park.
- 2024-01 – 2024-12 · **Visiting Student**, Shen Lab, Department of Biostatistics, Epidemiology and Informatics, Perelman School of Medicine, University of Pennsylvania (Penn), Philadelphia, USA, under the supervision of Prof. Li Shen.
- 2017-03 – 2017-12 · **Undergraduate Group Research Program (UGRP)**, BRAINLab, Division of Intelligent Robotics, Daegu Gyeongbuk Institute of Science & Technology (DGIST), under the supervision of Prof. Jinung An.
- 2017-01 – 2018-02 · **Undergraduate Research Assistant**, DEMRC, Department of Robotics and Mechatronics Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), under the supervision of Prof. Hongsoo Choi.

Education

- 2020-03 – 2026-02 · **Doctor of Philosophy**, Department of Robotics and Mechatronics Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, South Korea, under the supervision of Prof. Sang Hyun Park.
- 2015-03 – 2020-02 · **Bachelor of Engineering**, School of Undergraduate Studies, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, South Korea.

Research Publications

* indicates co-first authorship, and † indicates co-corresponding authorship.

In Submission

- **An, S.**, Kim, S., & Park, S. H. (2026a). Meta-Learning based EEG Foundation Model with Interpretable Functional Embeddings. In *Under Review*.

- Kim, S., Lee, K., **An, S.**, Chikontwe, M., Philip Kang, Adeli, E., Pohl, K. M., & Park, S. H. (2026). Semantic Context Guided Industrial Anomaly Detection with Semi-Supervised Few-Shot Part Segmentation. In *Under Review*.
- Lee, H., Lee, G., **An, S.**, & Park, S. H. (2026). CASTLE: Completion and Alignment Shape Transformer for Long-Bone Estimation. In *Under Review*.
- Lee, K.*, **An, S.***, Ryu, H., Park, H., & Park, S. H. (2026). Context-aware Large Language Model for Open-Vocabulary MEG-to-Text Decoding. In *Under Review*.

Journal Articles

- **An, S.**, Kang, M., Kim, S., Chikontwe, P., Shen, L., & Park, S. H. (2026). Subject-Adaptive Meta-Learning for Personalized BCI: A Fusion of Resting-State EEG Signal and Task-Specific Information. *Information Fusion (IF 15.5, JCR<2%)*.
- **An, S.**, Kim, J., Kim, S., Chikontwe, P., Jung, J., Jeon, H., & Park, S. H. (2024). Few shot Anomaly Detection using Positive Unlabeled Learning with Cycle Consistency and Co-occurrence Features. *Expert Systems with Applications (IF 7.5, JCR<10%)*.
- Ullah, I., **An, S.**, Chikontwe, P., Kang, M., Choi, J., & Park, S. H. (2024). Video Domain Adaptation for Semantic Segmentation using Perceptual Consistency Matching. *Neural Networks (IF 6.0, JCR<10%)*.
- **An, S.**, Kim, S., Chikontwe, P., & Park, S. H. (2023). Dual Attention Relation Network with Fine-Tuning for Few Shot EEG Motor Imagery Classification. *IEEE Transactions on Neural Networks and Learning Systems (TNNLS) (IF 10.4, JCR<5%)*.
- Kim, S., Chikontwe, P., **An, S.**, & Park, S. H. (2023). Uncertainty-aware semi-supervised few shot segmentation. *Pattern Recognition (IF 8, JCR<10%)*.
- Cha, H., **An, S.**, Choi, S., Yang, S., Park, S. H., & Park, S. (2022). Study on Intention Recognition and Sensory Feedback: Control of Robotic Prosthetic Hand Through EMG Classification and Proprioceptive Feedback Using Rule-based Haptic Device. *IEEE Transactions on Haptics*.

International Conferences

- **An, S.**, Kim, S., & Park, S. H. (2026b). Subject- and Task-Aware EEG Foundation Model. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- Kim, Y., Kim, S., **An, S.**, & Park, S. H. (2026). Rethinking RAG: Mitigating Text-Driven Hallucinations via Feature-Level Retrieval Integration in Medical LVLMs. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) (Early Accept<Top9%)*.
- Park, H., Kim, S., Arsalane, W., **An, S.†**, & Park, S. H.†. (2026). Retrieval-Guided Federated Semi-Supervised Learning for Medical Image Segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- Lee, S. *., Kim, S. *., **An, S.**, Lee, S. C. †., & Park, S. H. †. (2025). Few-Shot Logical Anomaly Detection with Text-based Logics via Language-Image Contrastive Training. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
- **An, S.**, Kang, M., Kim, S., Chikontwe, P., Shen, L., & Park, S. H. (2024). Subject-Adaptive Transfer Learning Using Resting State EEG Signals for Cross-Subject EEG Motor Imagery Classification. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) (Early Accept<Top11%)*.
- **An, S.**, Shen, L., & Park, S. H. (2024). Motor Imagery EEG Signal Generation from Resting-State Signal. In *Annual International Conference of the IEEE Engineering in Medicine & Biology Society Research Poster (EMBC Research Poster)*.
- Kim, S., **An, S.**, Chikontwe, P., Kang, M., Adeli, E., Pohl, K. M., & Park, S. H. (2024). Few Shot Part Segmentation Reveals Compositional Logic for Industrial Anomaly Detection. In *AAAI Conference on Artificial Intelligence (AAAI)*.

- Kim, S., **An**, **S.**, Chikontwe, P., & Park, S. H. (2021b). Bidirectional RNN-based few shot learning for 3d medical image segmentation. In *AAAI Conference on Artificial Intelligence (AAAI)*.
- Won, D. K., Jung, E., **An**, **S.**, Chikontwe, P., & Park, S. H. (2021). Low-Dose CT Denoising Using Pseudo-CT Image Pairs. In *MICCAI PRIME (International Workshop on PRedictive Intelligence In MEdicine)*.
- **An**, **S.**, Kim, S., Chikontwe, P., & Park, S. H. (2020). Few-Shot Relation Learning with Attention for EEG-based Motor Imagery Classification. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- Won, D. K., **An**, **S.**, Park, S. H., & Ye, D. H. (2020). Low-dose CT denoising using octave convolution with high and low frequency bands. In *MICCAI PRIME (International Workshop on PRedictive Intelligence In MEdicine)*.

Domestic Conferences

- **An**, **S.**, Park, H., Kim, S., & Par, S. H. (2026). Meta-Learning based EEG Foundation Model. In *Workshop on Image Processing and Image Understanding (IPIU)*.
- **An**, **S.**, Kang, M., & Park, S. H. (2025). Subject-adaptive meta-learning for EEG motor imagery classification. In *Workshop on Brain Engineering Society of Korea*.
- Ryu, H., **An**, **S.**, Kang, M., Kim, Y., & Park, S. H. (2025). Single-Scan Attentive Mamba for 3D Brain Tumor Segmentation. In *Workshop on Brain Engineering Society of Korea*.
- Lee, D., **An**, **S.**, & Park, S. H. (2023). Anomaly Detection on Abdominal CT and Brain MRI based on Densitiy-Estimation. In *Workshop on Image Processing and Image Understanding (IPIU)*.
- **An**, **S.**, Kim, S., Chikontwe, P., Park, H., Abdul, T., & Park, S. H. (2022). Dual Attention Relation Network for Few Shot EEG Motor Imagery Classification. In *Summer Annual Conference of IEIE*.
- Kim, S., **An**, **S.**, Chikontwe, P., & Park, S. H. (2021a). Few Shot Learning based on Bidirectional Recurrent Neural Network for 3D Medical Image Segmentation. In *Workshop on Image Processing and Image Understanding (IPIU)*.
- **An**, **S.**, Kim, S., & Park, S. H. (2020). Neural Spike Detection using Deep Convolutional Neural Network. In *Workshop on Image Processing and Image Understanding (IPIU)*.
- Lee, H., **An**, **S.**, Jeon, S., Kim, K., Lee, S.-G., Lee, S., . . . Choi, H. (2017). Development of Helical Drilling Microrobot Based on Catheter and Guidewire for Chronic Total Occlusion Treatment. In *The Korean Society of Mechanical Engineers Annual Meeting*.

Review Services

- Medical Image Analysis
- Neural Networks
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Industrial Informatics (TII)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- Journal of Biomedical and Health Informatics (JBHI)
- Cognitive Neurodynamics
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)
- International Concerence on Bioinformatics and Biomedicine (BIBM)
- The AAAI Conference on Artificial Intelligence (AAAI)

Awards

2026	·	Hyeseul Best Paper Award
2023	·	DGIST Outstanding Researcher Award
	·	DGIST Post-Graduate Research Abroad Awards
2021	·	Ranked 3rd in SNUH Sleep AI Challenge
	·	Paper Awards at ICROS 2021

Presentations

2026	·	IPIU 2026 Doctoral consortium

References

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